

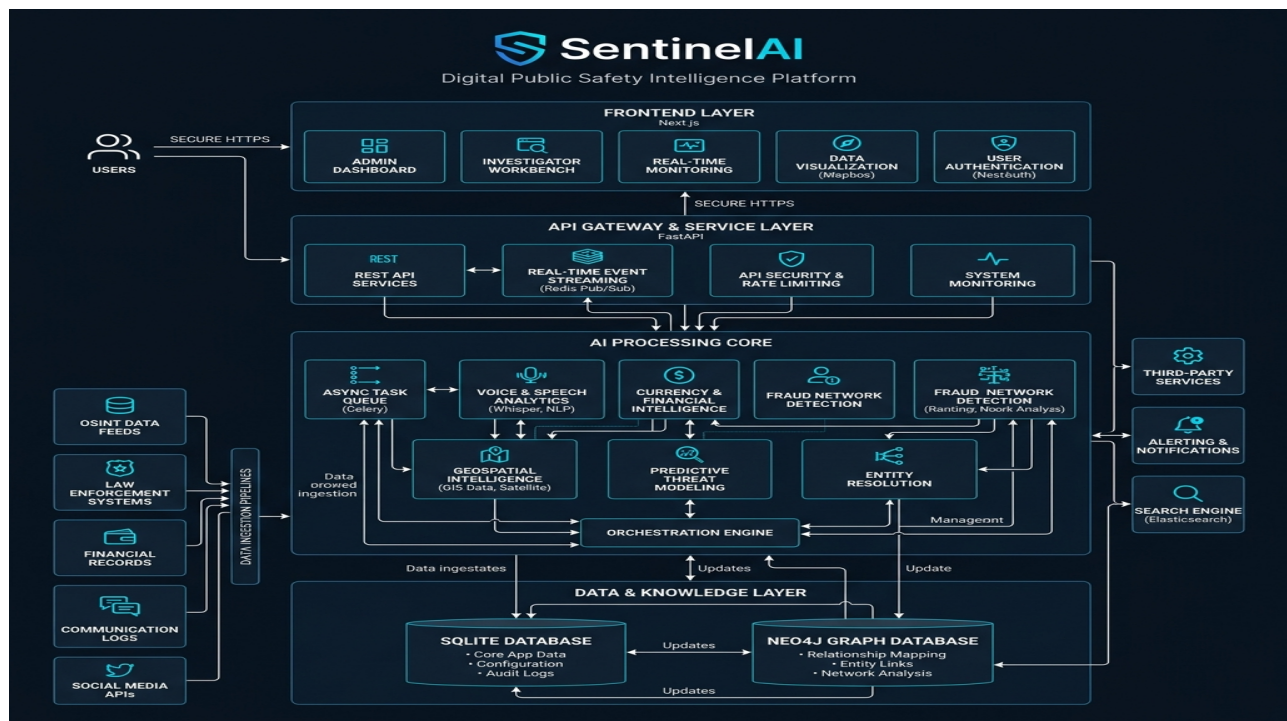
■ SentinelAI – Technical & Architectural Documentation

Welcome to the official technical documentation for **SentinelAI** (National Digital Public Safety Intelligence Platform). SentinelAI is developed as a production-grade, local-first monolithic platform designed for the **Smart India Hackathon 2024** under the problem statement **AI for Digital Public Safety**.

This document describes the codebase architecture, database models, integrated AI models, and user role mechanics.

■ System Architecture

SentinelAI utilizes an integrated, multi-role monolith structure designed to simulate instant coordination between citizens, local police, financial institutions, telecom carriers, and administrators.



■ Repository Directory Layout

The workspace repository organizes the monolith implementation cleanly between services:

```
ET AI/
■■■ .env.example           # Environment variables template for database ports
■■■ .gitignore             # Git excludes for caches, Next.js build files, and venv modules
■■■ package.json           # Root npm scripts to control both developer servers simultaneously
■■■ requirements.txt       # PyPI python modules list for FastAPI and AI models
■■■ database/
■   ■■■ schema.sql        # Core SQL Schemas (optimized for SQLite & PostgreSQL compatibility)
■   ■■■ seed_data.py      # Automatic seeding script for demo database records
■■■ docs/
■   ■■■ detailed_document.md # Detailed system architectural guide (Markdown)
■   ■■■ detailed_document.pdf # Detailed system architectural guide (PDF)
■■■ backend/
■   ■■■ main.py           # Main FastAPI server entry point (configures CORS & mounts routes)
■   ■■■ config.py         # Settings controller reading environment configurations
■   ■■■ database/
■       ■■■ connection.py # SQLAlchemy Connection manager and Session factory
■   ■■■ ai/
■       ■■■ counterfeit_detector.py # YOLOv11 & EfficientNet-B4 counterfeit checker
■       ■■■ fraud_graph.py   # Neo4j and Fallback Mock fraud rings visualizer
■       ■■■ geospatial.py   # H3 Hexagonal indexing & ST-GCN crime map predictions
■       ■■■ predictive.py    # XGBoost Transaction risk and emerging campaign analysis
■       ■■■ scam_detector.py # IndicBERT + RoBERTa voice/text scam classification
■   ■■■ routers/          # API router files
■       ■■■ auth.py, citizen.py, police.py, bank.py, telecom.py, admin.py, search.py
■   ■■■ schemas/          # Input/Output validation models (Pydantic v2)
■■■ frontend/
    ■■■ package.json      # Next.js scripts & packages
    ■■■ postcss.config.mjs # Tailwind PostCSS configuration
    ■■■ tsconfig.json     # TypeScript configuration
    ■■■ app/
        ■■■ layout.tsx    # Base metadata, navbar wrapper, and styling imports
        ■■■ page.tsx      # Dynamic Multi-Role Landing page (with optimized 2-row grids)
        ■■■ analytics/    # Platform-wide security overview statistics
        ■■■ dashboard/    # Role-based panels (citizen, police, bank, telecom, admin)
```

■■ Core Technology Stack

- **Frontend:** Next.js 16 (App Router, TSX) + Tailwind CSS v4 + Framer Motion (premium layout transitions) + Lucide Icons.
- **Backend:** FastAPI (Python 3.12+) + SQLAlchemy (ORM) + Pydantic v2 (Input validation).
- **Database:** SQLite (local single file [kavach.db](#) for serverless development) / PostgreSQL (for production).
- **Relationship Graph:** Neo4j (via bolt protocol [7687](#)) with a custom fallback mock engine.

■■ KAVACH Intelligent AI Modules

The backend AI logic simulates state-of-the-art architectures optimized for local host speed:

AI Module	Code File	Simulates Model Stack	Objective
KAVACH-VOICE	scam_detector.py	IndicBERT, Whisper v3, ECAPA-TDNN	Detects digital arrest, banking, utility, and courier scams in 12 Indian languages.
KAVACH-CURRENCY	counterfeit_detector.py	YOLOv11, EfficientNet-B4, ViT-B/16	Scans security thread, watermark, UV marks, and serial numbers.
KAVACH-NET	fraud_graph.py	GraphSAGE, GAT, Neo4j Bolt	Traces multi-hop transaction networks to find fraud ring endpoints.
KAVACH-GEO	geospatial.py	ST-GCN, Prophet, H3 Hexagons	Predicts crime hotspots and automatically suggests optimal patrol routes.
KAVACH-PREDICT	predictive.py	XGBoost Ensemble, Isolation Forest	Detects mule bank accounts, transaction velocities, and alerts target victims.

■ ■ Database Tables Schema

The SQLite schema in [kavach.db](#) structures information via the following main entities:

1. [users](#): Main system actors. Hashed with SHA-256 for secure mobile lookup, containing user role access variables.
2. [police_officers](#): Connected metadata for law enforcement users including badges, ranks, and police station locations.
3. [fraud_cases](#): FIR-style reported scams containing case status, risk levels, and automated case summaries.
4. [evidence](#): Digital evidence records uploaded by citizens (audio transcriptions, SMS texts, bill screenshots).
5. [fraud_actors](#): Tracked phone numbers, bank accounts, and IMEI codes suspected of hosting scam campaigns.
6. [geospatial_crime_events](#): Categorized coordinates tracked under H3 index cells for crime mapping.
7. [audit_logs](#): System logs tracking administrative and investigative actions.

■ API Reference (Endpoints)

All endpoints reside under [/api/v1](#) prefix:

- **Authentication** ([/auth](#)):

- **POST /register**: Registers new users with specialized roles.
- **POST /login**: Validates password and generates JWT access tokens.
- **POST /verify-otp**: Demo verification step using default code [123456](#).
- **Citizen (/citizen)**:
 - **GET /alerts**: Returns local security advisories.
 - **POST /report**: Uploads fraud evidence (text, audio, image).
- **Police (/police)**:
 - **GET /dashboard**: Overview statistics on active crime investigations.
 - **GET /cases**: Retrieves full list of active/resolved cases.
 - **GET /network-graph**: Feeds the interactive fraud ring visualizer.
- **Bank (/bank)**:
 - **GET /transactions/high-risk**: Yields suspicious accounts flagged for hold.
 - **POST /mule/freeze**: Freezes money mule accounts immediately.
- **Telecom (/telecom)**:
 - **GET /numbers/flagged**: Identifies numbers broadcasting spam voice/SMS.
 - **POST /numbers/block**: Simulates SIM disconnection and IMEI blacklisting.